

C035M DESIGN RULE MANUAL SUPPLEMENT FOR ANALOGUE OPTION (C035M-A)

Owner : Technology Engineering CMOS

Location : N.A.

De inlichtingen van dit document zijn eigendom van ALCATEL MICROELECTRONICS N.V. Oudenaarde en reproducties, toepassingen of mededelingen aan derden zijn streng verboden zonder schriftelijke toelating.

Les données contenues dans le présent document sont la propriété de ALCATEL MICROELECTRONICS S.A. Oudenaarde et leur reproduction, application ou communication à des tiers est strictement interdite sans accord écrit.

The contents of this document are the property of ALCATEL MICROELECTRONICS N.V. Oudenaarde and must not be copied, reproduced or disclosed to a third party without written consent.

REVISION STATUS SUMMARY

Revision	Requestor	Date	Request Number	Pages	Description
01	AL	30-11-1998	19565	01-30	New document
02	AL	15-12-1999	21686	01-34	Update for dummy metal, transistor matching, capacitor matching. Inclusion of new data for resistors

Alcatel Microelectronics	C035M Design rule manual supplement for analogue option (C035M-A)	Document DS13337	Revision 02	Date 15-12-1999	Page 1 Of 34
-----------------------------	--	---------------------	----------------	--------------------	-----------------

TABLE OF CONTENTS

1	CHANGES	4
1.1	CHANGES FROM REVISION 1 TO REVISION 2	4
2	PURPOSE	4
3	SCOPE	4
4	REFERENCE DOCUMENTS	5
5	PROCESS AND MASK DETAILS – C035M-A	6
5.1	CHARACTERISED DEVICES FOR CIRCUIT DESIGN	6
5.2	KEY FEATURES OF EXTRA ANALOGUE MODULES	6
5.3	SIMPLIFIED PROCESS FLOW C035M-A	7
5.4	COMPATIBILITY WITH C05M-A	10
6	LAYOUT RULES	10
6.1	INTRODUCTION	10
6.2	RULES FOR CAPACITORS	10
6.2.1	DEFINITIONS - CAPACITOR	10
6.2.2	POLYSILICON (IGS LAYER 13)	11
6.2.3	CAPA (IGS LAYER 12)	13
6.2.4	POLY2 (IGS LAYER 11)	14
6.2.5	SALICIDE PROTECT (IGS LAYER 18)	14
6.2.6	NLDDPROT (IGS LAYER 38)	15
6.2.7	NPLUSPROT (IGS LAYER 66)	15
6.2.8	METAL1 (IGS LAYER 23)	15
6.3	RULES FOR HIPO RESISTOR	17
6.3.1	DEFINITIONS - HIPO RESISTOR	17
6.3.2	ACTIVE AREA (IGS LAYER 2)	17
6.3.3	POLYSILICON (IGS LAYER 13)	18
6.3.4	HIPO (IGS LAYER 26)	19
6.3.5	P+ IMPLANT (IGS LAYER 17)	21
6.3.6	SALICIDE PROTECT (IGS LAYER 18)	21
6.3.7	EXTRA RULES FOR MATCHED HIPO RESISTORS	21
6.4	RELIABILITY LAYOUT RULES	25
6.4.1	HOT CARRIER INJECTION CONSTRAINTS FOR ANALOGUE APPLICATIONS	25

7 ELECTRICAL PARAMETERS	26
7.1 TRANSISTORS - GENERAL PROPERTIES	26
7.2 MATCHING OF TRANSISTORS	26
7.2.1 EXPERIMENTAL RESULTS ON MATCHING	27
7.3 RESISTORS	28
7.3.1 MEASUREMENT CONDITIONS FOR THE ELECTRICAL MODEL	28
7.3.2 LINEARITY vs BIAS MODEL AND MEASUREMENT CONDITIONS	28
7.3.3 LINEARITY vs TEMP. MODEL AND MEASUREMENT CONDITIONS	29
7.3.4 MEASUREMENT CONDITIONS FOR MATCHING	29
7.3.5 MODELLING FORMULA USED FOR MATCHING	29
7.4 DOUBLE POLY CAPACITOR	33
7.4.1 VOLTAGE LINEARITY	33
7.4.2 CAPACITOR MATCHING	34

1 CHANGES

1.1 Changes from revision 1 to revision 2

- change numbering of poly2 rules to line up with DRC.
- introduction of rules for metdis layer to be used to protect against generation of extra metal features on structures where matching is critical.
- update of hot carrier reliability table.
- update of MOS matching data.
- clarification of resistor size used for resistor linearity measurements.
- inclusion of parameters for n+ and p+ unsilicided poly resistors (matching, linearity, temperature coefficient)
- correction error in capacitor linearity formula
- correction of error in capacitor matching figures – line-up units and coefficients.

2 PURPOSE

This Design Rule Manual, together with the core digital C035M-D Design Rule Manual, provides the process sequence, layout rules and electrical parameters needed for the design, layout, processing and electrical characterisation of integrated circuits using the Alcatel Microelectronics C035M-A technology.

The C035M-A process is a double level poly, five level metal, twin-tub CMOS process using a < 100> p/p⁺ epitaxial substrate. It is based on the C035M-D digital process (see C035M-D Design Rule Manual DS13330). To this core digital process extra process modules have been added so as to provide capacitors and resistors required for analogue design.

The technology is developed for the design of analogue circuits operating at a power supply voltage of 3.3 Volt (+ 0.3/-0.6 Volt).

3 SCOPE

This document applies to Design Engineers, Technology Engineers, CAD Engineers, Process Engineers, Quality Assurance Engineers, Reliability Engineers and Product Engineers.

This specification provides information about the extra capacitor and high ohmic polysilicon resistor modules available in C035M-A and, additionally, some extra information about the active devices which is especially relevant for analogue design (eg matching).

The specification should be read in conjunction with the C035M-D Digital Design Rule Manual. These two documents are the prime reference to be applied for the

Alcatel Microelectronics	C035M Design rule manual supplement for analogue option (C035M-A)	Document DS13337	Revision 02	Date 15-12-1999	Page 4 Of 34
-----------------------------	--	---------------------	----------------	--------------------	-----------------

design and layout of any C035M-A product to be fabricated in Alcatel Microelectronics.

4 REFERENCE DOCUMENTS

- DS 13330 C035M-D Design Rule Manual (DRM)
- DS 13600 Assembly Layout Rules

Alcatel Microelectronics	C035M Design rule manual supplement for analogue option (C035M-A)	Document DS 13337	Revision 02	Date 15-12-1999	Page 5 Of 34
-----------------------------	--	----------------------	----------------	--------------------	-----------------

5 PROCESS AND MASK DETAILS – C035M-A

INTRODUCTION

The Alcatel Microelectronics 0.35 μm CMOS technology family consists of a core digital technology C035M-D and a mixed-signal technology C035M-A. C035M-A is built on C035M-D with the addition of two specific options: high precision double polysilicon capacitors and high ohmic polysilicon resistors.

The C035M technology family is designed for 3.3 volt (+ 0.3/-0.6 volt) power supply operation and features salicided junctions and polysilicon, and unsalicated resistors and I/O transistors.

5.1 Characterised devices for Circuit Design

The devices available in C035M-A are those given in the C035M-D with the addition of the following:

- Highly Linear poly-1/poly-2 capacitor ($1.1 \text{ fF}/\mu\text{m}^2$),
- High Ohmic p-type-Poly Resistor defined as HIPO Resistor ($1 \text{ k}\Omega/\text{sq}$).

Note that a distinction is made in the C035M-D Design Rule Manual between devices which are fully characterised, those which are authorised only for specific applications and those which are natural devices. It is strongly advised that analogue designs should be made using only the fully characterised devices.

5.2 Key Features of extra Analogue Modules

POLY/POLY CAPACITOR: Formed using the gate poly layer as bottom plate and a deposited dielectric and additional poly layer to complete the capacitor.

HIPO RESISTOR: High ohmic polysilicon resistor formed in the gate poly layer but using an additional masking step and separate implant to achieve the appropriate sheet resistance.

Alcatel Microelectronics	C035M Design rule manual supplement for analogue option (C035M-A)	Document DS13337	Revision 02	Date 15-12-1999	Page 6 Of 34
-----------------------------	--	---------------------	----------------	--------------------	-----------------

5.3 Simplified process flow C035M-A

This section gives a brief summary of the process flow for the C035M-A technology.

- Active Area
 - oxide, amorphous silicon, nitride stack deposition
 - active area patterning
 - field oxidation
 - sacrificial oxidation
- N-well and PMOS channel doping
 - n-well patterning
 - retrograde n-well implant
 - PMOS V_t & anti-punch through implant
 - annealing
- P-well and NMOS channel doping
 - p-well patterning
 - retrograde p-well implant
 - NMOS V_t & anti-punch through implant
- Gate
 - wet etch sacrificial oxide
 - gate oxidation
 - polysilicon deposition and doping
- Capacitor
 - masked implant capacitor bottom plate
- HIPO Resistor
 - HIPO mask
 - HIPO implant
- Capacitor formation, poly anneal & patterning
 - capacitor dielectric formation
 - capacitor top plate deposition and doping
 - polysilicon anneal
 - capacitor top plate & dielectric patterning
 - gate poly and capacitor bottom plate patterning
 - polysilicon oxidation
- Lightly Doped Drain's (LDD)
 - NMOS LDD mask & implant
 - PMOS LDD mask & implant
 - oxide sidewall spacer deposition
 - oxide sidewall spacer etch

Alcatel Microelectronics	C035M Design rule manual supplement for analogue option (C035M-A)	Document DS13337	Revision 02	Date 15-12-1999	Page 7 Of 34
-----------------------------	--	---------------------	----------------	--------------------	-----------------

- Source & Drain
 - pre-implant oxide
 - N⁺ source/drain mask & implant
 - P⁺ source/drain mask & implant
 - junction annealing
- Salicide
 - salicide protect deposition
 - salicide protect patterning
 - Titanium deposition
 - Titanium salicide formation
 - selective etch
 - salicide anneal
- Inter-Level Dielectric (ILD)
 - oxide deposition
 - doped oxide deposition
 - planarisation
 - oxide deposition and densification
- Contacts & metal 1
 - contact window patterning
 - glue layer deposition
 - tungsten plug formation
 - metal 1 deposition
 - metal 1 patterning
- Inter-Metal Dielectric (IMD1)
 - oxide deposition
 - CMP planarisation
- Via 1 & metal 2
 - via 1 patterning
 - glue layer deposition
 - tungsten plug formation
 - metal 2 deposition
 - metal 2 patterning
- Inter-Metal Dielectric (IMD2)
 - oxide deposition
 - CMP planarisation
- Via 2 & metal 3
 - via 2 patterning
 - glue layer deposition
 - tungsten plug formation
 - metal 3 deposition
 - metal 3 patterning
- Inter-Metal Dielectric (IMD3)
 - oxide deposition
 - CMP planarisation

Alcatel Microelectronics	C035M Design rule manual supplement for analogue option (C035M-A)	Document DS 13337	Revision 02	Date 15-12-1999	Page 8 Of 34
-----------------------------	--	----------------------	----------------	--------------------	-----------------

- Via 3 & metal 4
 - via 3 patterning
 - glue layer deposition
 - tungsten plug formation
 - metal 4 deposition
 - metal 4 patterning
- Inter-Metal Dielectric (IMD4)
 - oxide deposition
 - CMP planarisation
- Via 4 & metal 5
 - via 4 patterning
 - glue layer deposition
 - tungsten plug formation
 - metal 5 deposition
 - metal 5 patterning
- Final passivation and annealing

IGS LAYERS FOR C035M-A

The requirements for IGS numbering described in the C035M-D Design Rule Manual are also applicable to C035M-A.

In addition, some extra IGS layers are used in C035M-A. A list of these is given in the following table.

IGS layer Number	IGS layer Name	Mask Number	Drawn?	Mask Name
12	capa	85	yes	CAPA
26	hipo	95	yes	HIPO
11	poly2	60	yes	POLY2

Note also that the presence of analogue layers also affects the generation of some other layers, as follows:

(1) NLDD mask data = drawn pplus + drawn nliddprot
+ drawn capa + drawn hipo

(2) N⁺ IMPLANT mask data = drawn pplus + drawn nplusprot
+ drawn capa + drawn hipo

5.4 Compatibility with C05M-A

There is a high level of compatibility with C05M-A in terms of layout rules. In case it is required to port designs from this technology, advice should be sought from the Maskprep group regarding automated conversion of cells.

6 LAYOUT RULES

6.1 Introduction

Conventions for specifying layout rules and requirements for grid sizes, chip dimensions, product identification, mask identification, iteration numbering, CD structures, I/O's and diagonal layout rules are given in the C035M-D Design Rule Manual.

THESE REQUIREMENTS AND THE LAYOUT RULES FOR THE CORE DIGITAL PROCESS APPLY IN FULL TO C035M-A.

Note also that additional layout requirements apply to guarantee the models for matching of transistors, capacitors and resistors (see the appropriate sections of this manual).

6.2 Rules for capacitors

This section summarises the analogue-specific electrical design rules for the poly/poly capacitor or combined poly/poly capacitor and HIPO resistor options.

6.2.1 Definitions - Capacitor

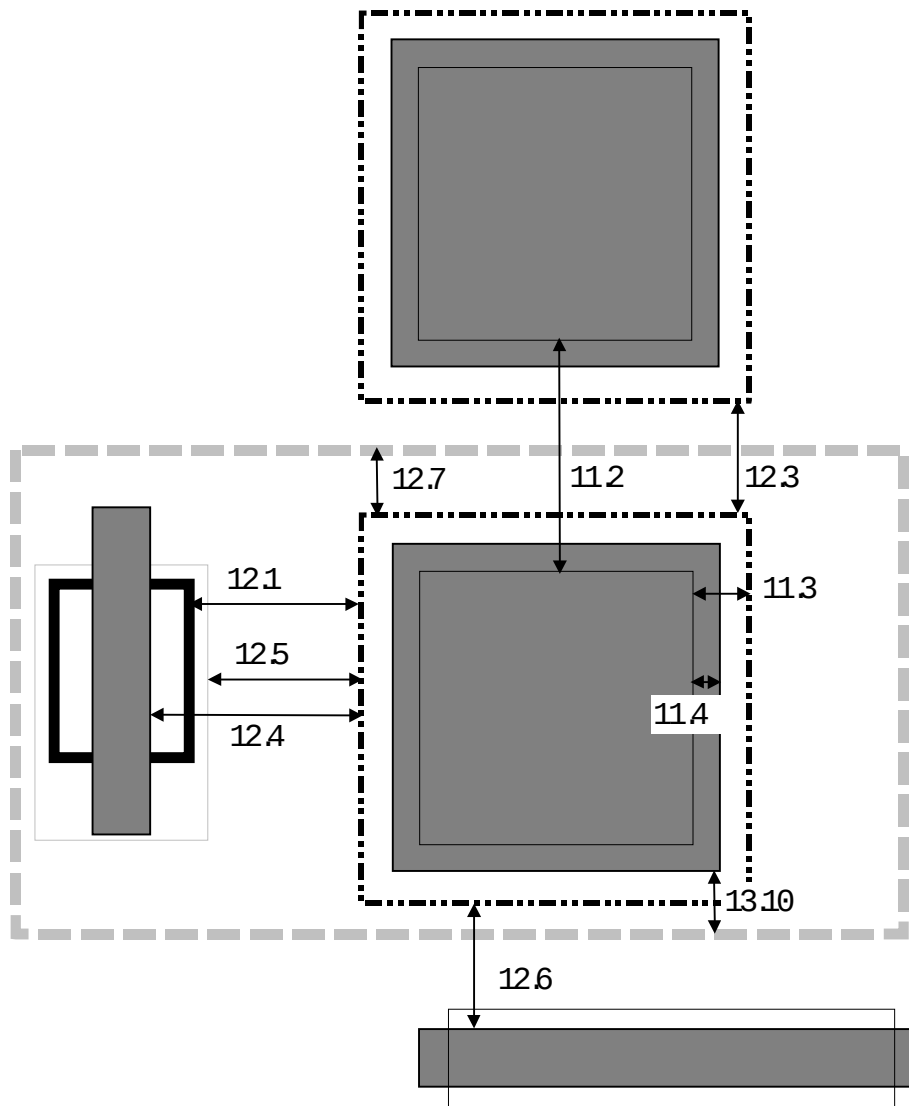
The Poly2/Poly1 Capacitor is formed by means of an oxide layer, sandwiched between heavily doped poly (CAPA layer is used as implant mask) and heavily doped Poly2. The figures in the following table are based on cross-sections and in-line thickness measurements. Values derived from electrical measurements can be different. See section 7 for definitive values of electrical parameters.

DESCRIPTION	MINIMUM [Å]	NOMINAL [Å]	MAXIMUM [Å]
Polycide 1 Thickness	2250	2500	2750
Poly 2 to Poly 1	270	300	330
Polycide 2 Thickness	2250	2500	2750

6.2.2 POLYSILICON (IGS layer 13)

If parasitic capacitance is a concern, Poly2/Poly1 capacitors can be placed on N-Well.

13.10	Minimum enclosure of polysilicon by n-well in case of Poly2/ Poly1 capacitor on n-well	1.0
-------	--	-----



N well



Active



Capa



Pplus



Poly



Poly2



Saprot

6.2.3 CAPA (IGS layer 12)

Inside feature is Poly2/Poly1 Capacitor bottom plate implant.

12.0	Capa is not allowed on active	-
12.1	Minimum capa spacing to active	0.4
12.2	Minimum capa width	1.0
12.3	Minimum capa spacing	1.0
12.4	Minimum capa distance to polysilicon	2.0
12.5	Minimum capa distance to pplus	0.8
12.6	Minimum capa distance to poly intersection with sal-prot of poly resistor	2.0
12.7	Minimum capa enclosure by n-well in case of Poly2/Poly1 capacitor on n-well	1.0

6.2.4 POLY2 (IGS layer 11)

Inside feature is Poly2/ Poly1 capacitor top plate.

11.0	Poly2 only allowed on field.AND.polysilicon.AND.capa	-
11.1	Minimum poly 2 width	1.2
11.2	Minimum poly 2 spacing	1.0
11.3	Minimum poly 2 enclosure by capa	1.0
11.4	Minimum poly 2 enclosure by polysilicon	1.0
11.5	Minimum poly 2 distance to contact	0.6
11.6	Minimum poly 2 enclosure of contact	0.6
11.9	Minimum poly 2 enclosure by sal-prot	0.6
11.10	Minimum poly 2 distance to contact on active for contact between two, three or four poly2 regions	3.5

6.2.5 SALICIDE PROTECT (IGS layer 18)

18.30	Minimum sal-prot distance to contact on poly 2	0.3
-------	--	-----

6.2.6 NLDDPROT (IGS layer 38)

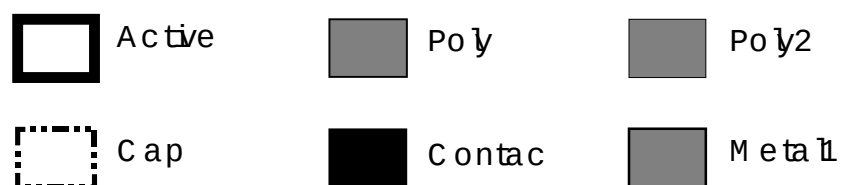
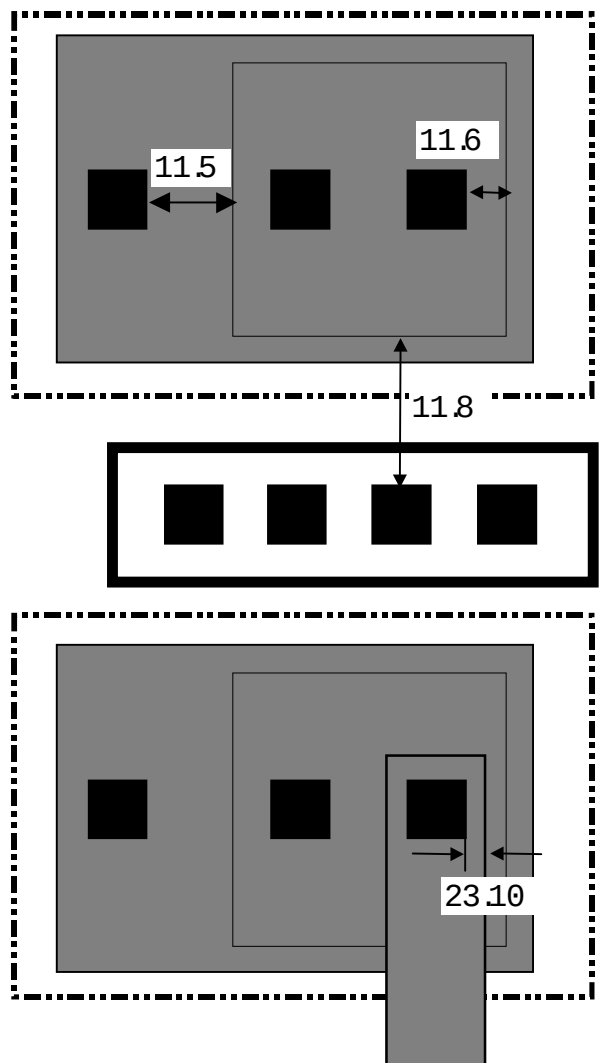
38.1	Minimum nlddprot distance to capa	0.5
38.2	Nlddprot and capa can be merged together if necessary	

6.2.7 NPLUSPROT (IGS layer 66)

38.1	Minimum nplusprot distance to capa	0.8
	If less, must be	0.0

6.2.8 METAL1 (IGS layer 23)

23.10	Minimum metal 1 enclosure of contact on poly 2	0.25
-------	--	------



6.3 Rules for HIPO RESISTOR

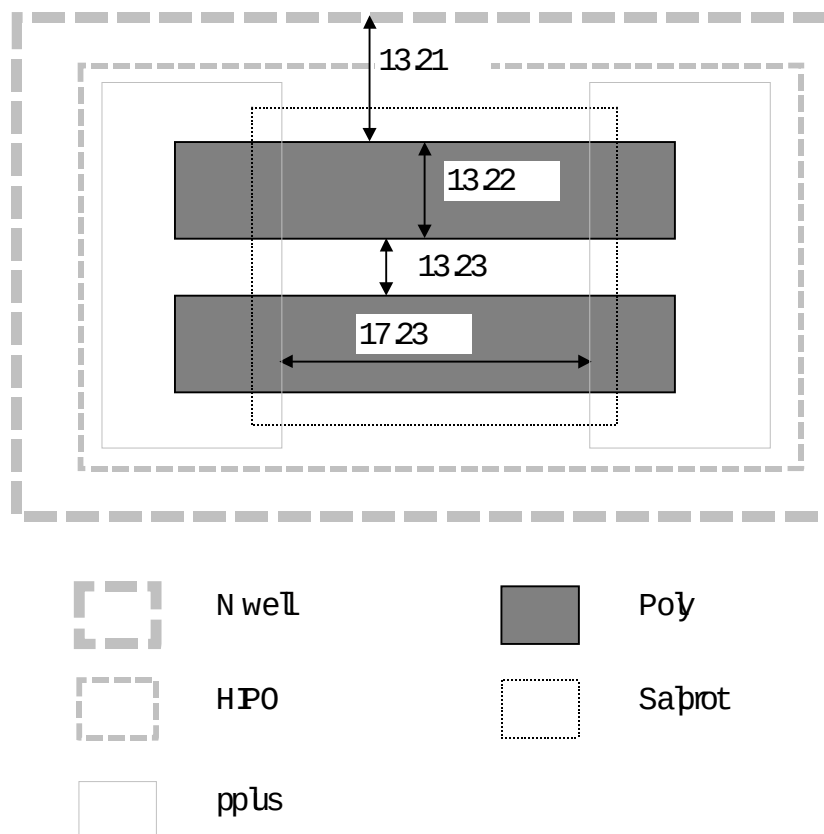
The HIPO resistor is an unsalicated poly resistor.

6.3.1 Definitions - HIPO RESISTOR

- HIPO resistor : Unsalicated poly area doped P- type between two heavily doped p+ type poly regions defining the heads of the HIPO resistor.
(hipo resistor= poly.AND.sal-prot.AND.hipo.AND.(.NOT.pplus)
- The resistance is defined by layer hipo.
- The drawn width is defined by polysilicon width.
- The drawn length is defined by the spacing between pplus.
- The resistor is automatically protected against N+ and NLDD implants (see section 5.3).
- HIPO dummy resistor : HIPO resistor whose heads are not contacted.
These resistors are drawn beside "active" resistor to improve matching performance.

6.3.2 ACTIVE AREA (IGS layer 2)

2.10	HIPO resistor is not allowed on active	-
------	--	---

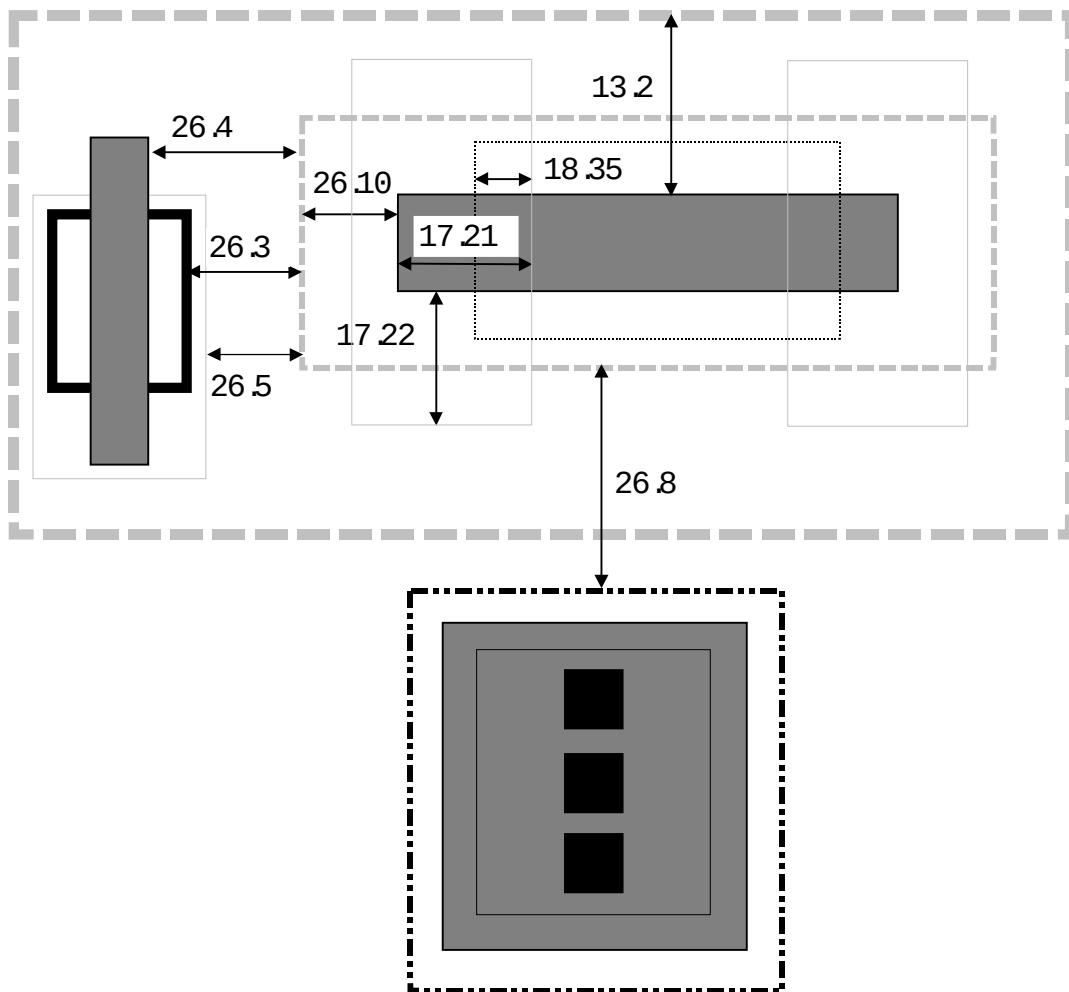


6.3.3 POLYSILICON (IGS layer 13)

13.20	HIPO resistor poly must be placed on n-well	-
13.21	Minimum HIPO resistor polysilicon enclosure by N-Well	1.5
13.22	Minimum width of polysilicon of HIPO resistor	1.2
13.23	Minimum spacing of polysilicon of HIPO resistor	0.8

6.3.4 HIPO (IGS layer 26)

26.1	Minimum hipo width	0.8
26.2	Minimum hipo spacing	0.8
26.3	Minimum hipo distance to active	0.5
26.4	Minimum hipo distance to poly	1.0
26.5	Minimum hipo distance to pplus	0.8
26.6	Minimum hipo distance to nllddprot If less, must be	0.8 0.0
26.7	Minimum hipo distance to nplusprot If less, must be	0.8 0.0
26.8	Minimum hipo distance to capa	0.8
26.9	Polysilicon of HIPO resistor crossing hipo boundary is not allowed	-
26.10	Minimum hipo enclosure of polysilicon of HIPO resistor	1.6



N well



Active



Poly



Cap



Poly2



HPO



ppls



Saprot

6.3.5 P+ IMPLANT (IGS layer 17)

17.20	Pplus must be placed on the heads of the HIPO resistor	-
17.21	Minimum pplus intersection with polysilicon of HIPO resistor	1.45
17.22	Minimum pplus extension on polysilicon of HIPO resistor	0.5
13.23	Minimum length of HIPO resistor (i.e. separation of pplus regions)	5.0

6.3.6 SALICIDE PROTECT (IGS layer 18)

18.35	Minimum sal-prot intersection with pplus on heads of HIPO resistors	0.5
18.36	Unsalicided poly crossing pplus boundary is allowed in hipo (in this case the normal rule 18.14 in C035M-D does not apply)	-

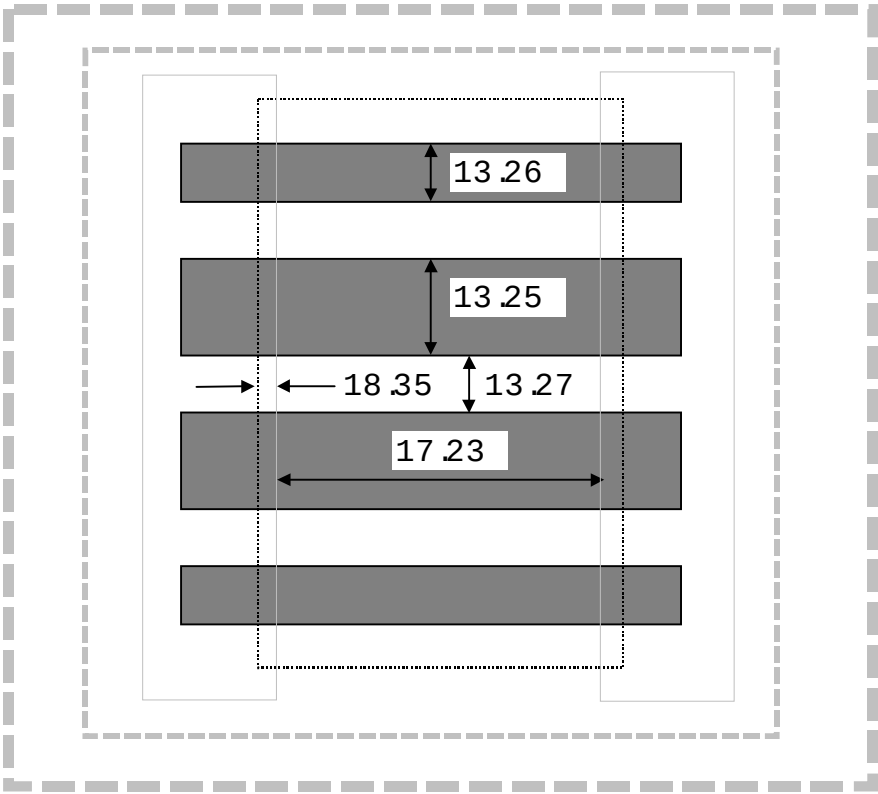
6.3.7 Extra rules for matched HIPO RESISTORS

- The rules summarised below are NOT checked with DRC
- All resistors must have the same width
- All resistors should have the same length
- All resistors must be in parallel with same length
- Dummy resistors must be used with the same space as the matched resistors.

13.25	Minimum width of HIPO resistor polysilicon for matched resistor	8.0
13.26	Minimum polysilicon width for dummy HIPO resistors	4.0
13.27	Polysilicon spacing (min = max) for matched HIPO resistors	1.5

17.23	Minimum HIPO resistor length if all matched resistors have the same length	35.0
17.24	Minimum HIPO resistor length if matched resistors do not all have the same length	200
26.11	HIPO enclosure of POLY of HIPO resistor	2.5
23.11	METAL1 over polysilicon of HIPO resistor is not allowed [†]	-
23.12	Minimum metal 1 distance to HIPO resistor (except on contact head)	2.0
23.13	Minimum metal 1 distance to related hipo resistor	0.7
27.5	Metal 2 over HIPO resistor is not allowed [†]	-
27.6	Minimum metal 2 distance to HIPO resistor	2.5
34.5	Metal 3 over HIPO resistor is not allowed [†]	-
34.6	Minimum metal 3 distance to HIPO resistor	2.5
36.5	Metal 4 over HIPO resistor is not allowed [†]	-
36.6	Minimum metal 4 distance to HIPO resistor	2.5
53.5	Metal 5 over HIPO resistor is not allowed [†]	-
53.6	Minimum metal 5 distance to HIPO resistor	2.5

[†] In order to guarantee the matching figures specified in section 7, it is necessary to cover matched resistors with layer metdis so as to identify them and prevent the generation of dummy metal structures on top (see C035M-D DRM DS13330).



N well



Poly



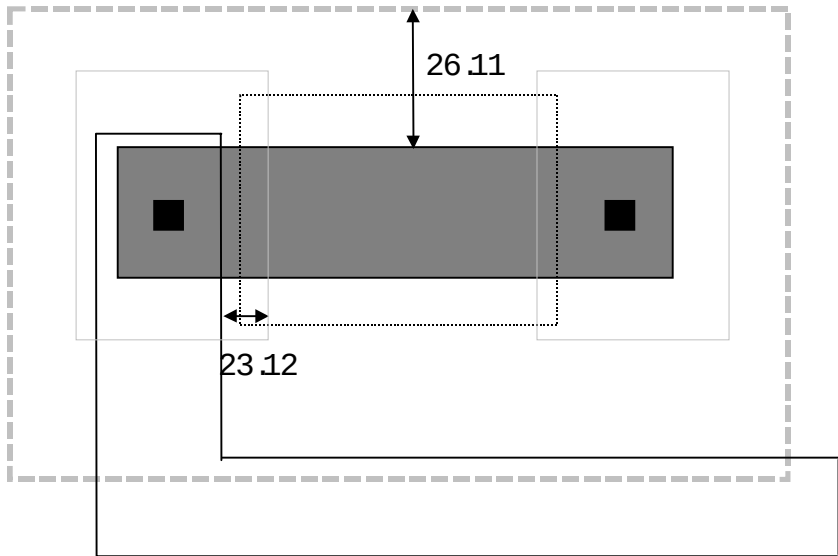
HPO



Salprot



ppus



Poly



HPO



Saprot



Contact



Metal1



ppus

6.4 Reliability layout rules

6.4.1 Hot carrier injection constraints for Analogue Applications

In the case of analogue circuits where different parameter sensitivities and different bias schematics may be found compared to the logic circuits and for other technical reasons (e.g. matching), analogue transistors are designed with channel length larger than minimum value. This strongly reduces hot carrier induced device performance degradation.

In order to guarantee a drift time of ≥ 10 years for either gm or I_{dlin} , the minimum gate-length needs to be adjusted.

Table 1, which is based on DC measurements of I_{dlin} drift (worst case), should help designers to choose a minimum gate length depending on the operation (V_{ds}/V_{gs}) conditions:

V_{gs} (V)	0.7 to 1	1.1	1.2	1.3	1.4 to 1.5	1.6 to 1.7	1.8 to 2.2	2.3 to 2.4	2.5 to 2.6	2.7 to 3.4	3.5	3.63
V_{ds} (V)												
2.5 to 2.8	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35
2.9	0.45	0.45	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35
3	0.5	0.5	0.5	0.5	0.4	0.35	0.35	0.35	0.35	0.35	0.35	0.35
3.1	0.5	0.5	0.5	0.5	0.5	0.5	0.35	0.35	0.35	0.35	0.35	0.35
3.2	0.7	0.5	0.5	0.5	0.55	0.5	0.4	0.35	0.35	0.35	0.35	0.35
3.3	0.9	0.9	0.9	0.7	0.7	0.5	0.5	0.35	0.35	0.35	0.35	0.35
3.4	1.05	0.9	0.9	0.7	0.7	0.5	0.5	0.5	0.35	0.35	0.35	0.35
3.5	1.15	1.1	1.1	1.05	0.9	0.7	0.5	0.5	0.5	0.4	0.35	0.35
3.63	1.3	1.4	1.4	1.3	1.05	0.9	0.7	0.7	0.5	0.5	0.5	0.45

Table 1

Summary of minimum gatelength [μm] for analogue applications as a function of V_{ds} and V_{gs} to have a I_{dlin} lifetime > 10 years. The minimum gate length for PMOS = 0.35 μm in all conditions.

7 ELECTRICAL PARAMETERS

All measurements are made at room temperature (300K) unless otherwise stated.

7.1 Transistors - general properties

For basic transistor parameters refer to specification DS 13330, "C035M-D Design Rule Manual" or to the SPICE model.

7.2 Matching of transistors

INTRODUCTION

Current mismatch observed between identically designed MOS transistors is linked to the statistical variations of the MOS characteristic parameters, which are mainly the threshold voltage V_T and the current factor β .

Generally, matching in a parameter P ($= I_D$, V_T or β) is defined as the standard deviation ($= 1\sigma$) of the distribution of the differences of P between the matched devices. The mean of the parameter differences is supposed to be zero and independent of distance, implying that there is no systematic offset nor gradient. For this reason, the values of the matching parameters given in the next section are valid only for designs respecting the following conditions

- identical design
- identical orientation
- identical direction of the current from source to drain
- distance between matched transistors $< 100\mu\text{m}$
- identical surroundings (equal distance to field, to capacitors, to metal lines etc.)
- no cross-over by unrelated metal (metals 1 to 5). In the case of metal coverage, a degradation-in matching is observed which is most severe for metal 1 coverage and least severe for metal 5 coverage. Note that, as described in the C035M-D Design Rule Manual, extra (dummy) metal features will be generated on all 5 metal layers. This can lead to a degradation in the matching performance. The matching models given below are only valid in the case that this dummy metal generation is blocked in the areas where transistor matching is required. In order to block generation of dummy metal, the metdis layer (63) must be used. Rules for the use of this in conjunction with matched transistors are shown below:

63.10	Minimum metdis enclosure of matched resistor pair	5
63.11	Minimum metdis enclosure of matched transistor gate	5

It is strongly advised to keep the use of metdis to a minimum. Large areas of unequal metal density inevitably lead to a loss of planarisation uniformity, which can influence parasitic capacitances and/or yield.

Due to plasma induced charge degradation it is greatly advised to

- protect the gate of the matched pairs by a diode,
- take care of the symmetry of the metal routing to the transistors.

7.2.1 Experimental results on matching

The model used for the gate voltage dependence of the current matching is

$$\sigma^2 \left(\frac{\Delta I_D}{I_D} \right) = \sigma^2 \left(\frac{\Delta \beta}{\beta} \right) + \frac{4 \cdot \sigma^2(\Delta V_T)}{(V_{gs} - V_T)^2}$$

valid in the strong inversion region ($V_{gs} - V_T > 0.25V$).

This equation applies for devices with $L \geq 2\mu\text{m}$ and $WL < 600\mu\text{m}^2$). For shorter channels, the matching behaviour of MOS transistors is deteriorated by extra mismatch causes such as CD control, source series resistance and short/narrow channel effects. It is therefore strongly advised not to use minimum channel lengths for matched MOS transistors. For transistors with area greater than $600\mu\text{m}^2$, no matching data are available; however, the possibility exists of some degradation due to longer distance non-uniformity.

Under the above conditions, the matching parameters $\sigma^2(\Delta V_T)$ and $\sigma^2(\Delta \beta / \beta)$ have the following dependence on transistor width (W) and length (L):

$$\sigma^2 = \frac{A^2}{WL} + B^2$$

The coefficients A^2 for $\sigma^2(\Delta V_T)$ and $\sigma^2(\Delta \beta / \beta)$ are given in the tables below for NMOS and PMOS for $V_{DS} = 3.3V$ and bulk bias $V_{BS} = 0V$. Due to the dependence of the matching on the specific measurement conditions used and the limited amount of data used to derive these figures, it is advised to use a 4-5 σ approach for the final calculation.

Parameter	Unit	NMOS	PMOS
A^2_{Vt}	$\text{mV}^2\mu\text{m}^2$	130	80
A^2_{β}	$\%\mu\text{m}^2$	6	6
B^2_{Vt}	mV^2	0	0
B^2_{β}	$\%^2$	0.04	0.1

7.3 Resistors

The model for the resistors uses the following equations:

$$R_0(T_{res}) = \rho_{SH} \frac{L - \Delta L}{W - \Delta W} + 2R_{HEAD}$$

with:

$$R_{HEAD} = \frac{R_C}{N_C} + \frac{R_{HU}(L_{HU} + \Delta L / 2) + R_{HS}L_{HS} + R_{HI}}{W - \Delta W}$$

W	:	Body Width [μm]
L	:	Body Length [μm]
T _{REF}	:	Reference Temperature (300 K)
ρ_{SH}	:	Sheet Resistance [Ohm/square]
ΔL	:	Drawn Length - Effective Length [μm]
ΔW	:	Drawn Width - Effective Width [μm]
R _{HS}	:	Salicided Sheet Resistance [Ohm/square]
L _{HS}	:	Salicided Head Length [μm] between contacts and body resistance
R _{HI}	:	Access Resistance between head and body [Ohm. μm]
R _{HU}	:	Unsalicided head sheet resistance [Ohm/square]
L _{HU}	:	Length of unsalicided head resistances [μm]
R _C	:	Contact Resistance [Ohm]
N _C	:	Number of Contacts

The parameters R_{HU} and L_{HU} , ΔL are only used for the HIPO resistor. However, they are given here in order to present the general model.

7.3.1 MEASUREMENT CONDITIONS FOR THE ELECTRICAL MODEL

The characterisation has been performed using W, L arrays with sizes ranging from W = 0.5 to 10 μm and L = 5 to 200 μm

7.3.2 LINEARITY vs BIAS MODEL AND MEASUREMENT CONDITIONS

The HIPO linearity has been characterised as a function of bias with the n-well under the resistors fixed at the same potential as one end of the resistor.

The voltage dependence of the HIPO resistors is fitted to the following equation:

$$R(V) = R_0 [1 + V_{cl}(V) + V_{cq}(V)^2]$$

Alcatel Microelectronics	C035M Design rule manual supplement for analogue option (C035M-A)	Document DS13337	Revision 02	Date 15-12-1999	Page 28 Of 34
-----------------------------	--	---------------------	----------------	--------------------	------------------

where V is the voltage across the hipo.

The resistor linearity has been measured on one resistor size per type. See the appropriate parameter table for further details.

7.3.3 LINEARITY vs TEMP. MODEL AND MEASUREMENT CONDITIONS

LINEARITY vs TEMPERATURE :

$$R(T) = R_0(T_{REF}) \left(1 + T_{CL}(T - T_{REF}) + T_{CQ}(T - T_{REF})^2 \right)$$

NOTE: Measurements have been performed at temperatures in the range -50 - + 150°C.

7.3.4 Measurement conditions for matching

The values given in the following tables have been measured on paired resistors surrounded by dummy resistors (= unconnected resistor on each side).

The distance between two matched resistors and between each resistor and the related dummy has been varied in the range 0.7-1.5 µm without a major impact on matching.

No metal routing is drawn on top of the pair.

7.3.5 MODELLING FORMULA USED FOR MATCHING

Modelling formula used for resistor matching:

$$\sigma^2 \left(\frac{\Delta R}{R} \right) = \frac{A^2}{(W - \Delta W)(L - \Delta L)}$$

NOTE: The above inverse area law is valid only for a certain family of dimensions (large areas). The minimum area used in the fitting is given below:

	Min. W.L used for $\frac{\Delta R}{R}$ law
HIPO, N+ & P+ unsilicided poly resistor	9 µm ²

For paired resistors with lower W.L area, the matching is strongly degraded, and the simple 1/√(WL) law is not usable.

HIPO RESISTOR PARAMETERS

PARAMETERS	MINIMUM (RES)	TYPICAL (RES)	MAXIMUM (RES)
ρ_{SH} [Ohm/sq]	800	1000	1200
ΔW [μm]	-	0.05	-
ΔL [μm]	-	0.5	-
R_{HS} [Ohm. μm]	1.5	2.5	5.0
R_{HI} [Ohm. μm]		45	
R_C [Ohm]	3.0	6.5	13.0
R_{HU} [Ohm]		110	
L_{HU} [μm]	-	0.5	-
V_{cl} [ppm/V] W/L= 1.8/20		-150	
V_{cq} [ppm/V ²] W/L= 1.8/20		-450	
T_{CL} [K ⁻¹]	-	-9E-4	-
T_{CQ} [K ⁻²]	-	3E-6	-
Matching Coefficient A [% μm]	-	7.5	-

P+ UNSILICIDED POLY RESISTOR PARAMETERS

PARAMETERS	MINIMUM (RES)	TYPICAL (RES)	MAXIMUM (RES)
ρ_{SH} [Ohm/sq]	80	120	160
ΔW [μm]	-	0.04	-
R_{HS} [Ohm. μm]	1.5	2.5	5.0
R_{HI} [Ohm. μm]			
R_C [Ohm]	3.0	6.5	13.0
V_d [ppm/V] W/L= 1.8/20		-50	
V_{cq} [ppm/V ²] W/L= 1.8/20		2200	
T_{CL} [K ⁻¹]	-	-8E-4	-
T_{CQ} [K ⁻²]	-	5E-7	-
Matching Coefficient A [% μm]	-	5.0	-

N+ UNSILICIDED POLY RESISTOR PARAMETERS

PARAMETERS	MINIMUM (RES)	TYPICAL (RES)	MAXIMUM (RES)
ρ_{SH} [Ohm/sq]	110	150	190
ΔW [μm]	-	0.09	-
R_{HS} [Ohm. μm]	1.5	2.5	5.0
R_{HI} [Ohm. μm]		70	
R_C [Ohm]	3.0	6.5	13.0
V_d [ppm/V] W/L= 1.8/20		100	
V_{cq} [ppm/V ²] W/L= 1.8/20		-2500	
T_{CL} [K ⁻¹]	-	-1E-3	-
T_{CQ} [K ⁻²]	-	2E-6	-
Matching Coefficient A [% μm]	-	10.5	-

7.4 Double Poly Capacitor

The specific capacitance of the POLY 1/POLY 2 capacitor is:

$$1.1 \pm 0.15 \text{ fF}/\mu\text{m}^2$$

In the C05M-A process, the voltage linearity of the capacitance is depending on whether the upper poly 2 plate of the capacitor is salicided or not. In the C035M-A process, the poly 2 implant dose is optimised to enhance the linearity of the salicided capacitor and the figures below relate to this device.

7.4.1 VOLTAGE LINEARITY

Assuming a quadratic voltage dependence:

$$C[V] = C_0 (1 + V_{CL} V + V_{CQ} V^2)$$

the linearity of the capacitor has been measured between -7 V and + 7 V (applied to the poly 2 plate) and values of linear and quadratic coefficients and are given in the table below.

	Min	Typical	Max
V_{CL} (ppm/V)	-40	2	40
V_{CQ} (ppm/V ²)	-6		

7.4.2 Capacitor Matching

Capacitor matching is modelled as follows:

$$\sigma^2\left(\frac{\Delta C}{C}\right) = \frac{A^2}{WL} + \frac{B^2}{(WL)^{\frac{3}{2}}}$$

In this model, the dimensions W, L are defined by the POLY 2 top plate. The coefficients A² and B² are given in the table below.

Parameter	Unit	Value
A ²	[% ² μm ²]	0.16
B ²	[% ² μm ³]	3

NOTES

- In the test structure used, the maximum area considered is W*L = 6400μm². For areas larger than this, the model given above cannot be guaranteed.
- The above model was derived from measurements on capacitors in close proximity to each other. The validity of the model cannot be guaranteed if this is not respected.